

Mini Web App on K3s - Architecture and Deployment Report

Title: Mini Web App K3s Deployment and Architecture

1. Overview

This project demonstrates a lightweight web application deployed on **K3s**, a lightweight Kubernetes distribution. The application consists of a **frontend** (Nginx), a **backend API** (Flask), and a **local registry** for hosting Docker images.

2. Architecture Diagram

```
flowchart TD
    subgraph User_Browser
        U[Browser] -->|http://192.168.56.104:30081| F SVC[frontend-service\nType: NodePort\nPort: 80\nNodePort: 30081]
    end

    F SVC --> FP1[frontend Pod 1\nNginx\nPort 80]
    F SVC --> FP2[frontend Pod 2\nNginx\nPort 80]

    FP1 --> HTML[index.html]
    FP2 --> HTML

    HTML -->|fetch('/api')| API REQ[/api request]

    API REQ ==> B SVC[backend-service\nType: ClusterIP\nPort: 5000]

    B SVC --> BP1[backend Pod 1\nFlask app.py\nPort 5000]
    B SVC --> BP2[backend Pod 2\nFlask app.py\nPort 5000]

    subgraph Registry
        R SVC[registry Service\nType: NodePort\nPort: 5000\nNodePort: 30500] --> R POD[registry Pod\nimage: registry:2\nPort 5000]
    end

    K3S[K3s + containerd] -->|pull images| R SVC
    K3S -->|pull frontend image| FP1
    K3S -->|pull frontend image| FP2
    K3S -->|pull backend image| BP1
    K3S -->|pull backend image| BP2
```

```
style User_Browser fill:#f0f8ff,stroke:#333,stroke-width:1px
style Registry fill:#ffe4b5,stroke:#333,stroke-width:1px
style K3S fill:#e0ffe0,stroke:#333,stroke-width:1px
```

3. Steps We Followed

1. **Setup K3s** on the VM with proper networking:
2. NAT adapter for internet access
3. Host-only adapter for host-VM communication
4. **Created backend (Flask) and frontend (Nginx) Dockerfiles:**
5. Backend listens on port 5000
6. Frontend listens on port 80 and proxies `/api` requests to backend-service
7. **Created local registry inside K3s** on NodePort 30500 for hosting Docker images.
8. **Configured DNS** in the VM:
9. Fixed `Temporary failure in name resolution`
10. Added `DNS=8.8.8.8 1.1.1.1` and `FallbackDNS=8.8.4.4` in `/etc/systemd/resolved.conf`
11. **Pulled base images** (`nginx:alpine`, `python:3.11-alpine`) successfully.
12. **Built and pushed frontend/backend images** to local registry:
13. Tagged with `192.168.56.104:30500/frontend:1.1` and `backend:1.1`
14. Configured Docker and K3s containerd to allow insecure HTTP registry
15. **Fixed backend deployment YAML** indentation issues to resolve `error converting YAML to JSON`.
16. **Applied deployments** and ensured pods became `Running`:
17. Verified frontend served `index.html`
18. Verified `/api` returned `Hello from Flask backend!`
19. **Cleaned up old terminating pods** after redeployments.

4. Challenges Faced

1. DNS Resolution Failure:

- Initially, `ping google.com` failed inside the VM
- Solved by updating `/etc/systemd/resolved.conf` and restarting `systemd-resolved`

4. HTTP Registry Issue:

- Push to `192.168.56.104:30500` failed due to HTTPS expectation
- Solved by adding insecure registry config for Docker and K3s containerd

7. YAML Indentation Error:

- `backend deployment` failed with `yaml: line 16: did not find expected key`
- Fixed proper 2-space indentation under `containers:` and `image:`

10. ImagePullBackOff / ErrImagePull:

- Occurred before configuring K3s to use HTTP registry
- Resolved after rebuilding, pushing, and configuring `registries.yaml`

13. Frontend Proxy:

- Browser `fetch('/api')` would fail until Nginx config was added to proxy `/api` to backend-service

5. Final Verification

- **Frontend access:** `http://192.168.56.104:30081`
- **Backend API via frontend:** `http://192.168.56.104:30081/api` → `{"message": "Hello from Flask backend!"}`
- **Pods status:** all frontend and backend pods `Running`
- **Registry access:** `curl http://192.168.56.104:30500/v2/` → `{}`

6. Conclusion

The mini web app is now fully deployed on K3s, with:

- Frontend Nginx pods accessible via NodePort
- Backend Flask pods accessible via ClusterIP service
- Local HTTP registry hosting images

- Proper DNS and registry configuration
- All previous issues resolved, including ImagePullBackOff and YAML syntax errors.